Problem Statement

ResearchLens AI - Evidence-grounded Research Agent

Selected problem	No. 1 - Research Agent
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01 Selected challenge

The supplied problem statement describes an AI research assistant that searches literature, summarizes papers, organizes references, understands research questions, retrieves relevant information, generates reports, suggests hypotheses and drafts research-paper sections. It also identifies citation management and data extraction as repetitive tasks to automate. [1]

Mandatory technology: “Use of IBM cloud lite services /IBM Granite is mandatory.” [1]

02 Project-specific problem

Students, researchers and project teams must combine information scattered across research PDFs and scholarly records. Methods, sample sizes, performance metrics and limitations can be difficult to locate and compare. Generated summaries can also confuse a paper’s own findings with results it cites, omit important context or attach an unsuitable citation.

ResearchLens AI addresses this need by keeping selected-paper scope, source passages, page references and review warnings alongside generated drafts. The goal is practical research assistance with inspectable evidence, not autonomous scientific validation.

03 Objective and intended users

Objective: Build a Python research-assistance prototype using IBM Granite through watsonx.ai and retrieval-augmented generation (RAG) to help users discover papers, inspect evidence, organize findings and prepare source-linked drafts.

Users: Students preparing projects, academic researchers reviewing literature and research teams comparing studies. The plant-disease papers used in the demonstration are test documents; the application is not itself a plant-disease classifier.

04 Expected outcomes

A usable research workspace; document-scoped answers and summaries; six-field extraction; two- or three-paper comparisons; research briefs, untested hypotheses and section drafts; and reference exports with visible evidence and uncertainty. These are intended capabilities, not a guarantee of complete or correct scientific answers.

[1] Source: IBM Skills Build for University Engagements, AICTE-2026, “Problem Statements on Agentic AI,” Problem Statement No. 1, page 1. This document adapts that challenge to the submitted ResearchLens AI implementation.

Implementation scope

A bounded, user-triggered workflow with IBM-hosted synthesis.

05 Proposed solution

Users upload permitted research PDFs or discover scholarly records through arXiv and Crossref. PDF text is extracted into page-aware passages. TF-IDF retrieval selects evidence from the chosen papers; IBM Granite generates a draft; deterministic checks inspect citation identifiers, numerical consistency and selected-source scope. The interface displays the draft, review status, warnings, evidence and export controls.

Search role	Plan a query and retrieve scholarly metadata or available abstracts.
Read role	Assemble relevant, bounded evidence from the selected sources.
Write role	Use IBM Granite via watsonx.ai for a bounded synthesis request.
Verify role	Check source links and consistency; flag issues for human review.

06 Functional and technical scope

Seven task routes: Research question answering, paper summarization, structured extraction (objective, method, sample, findings, metrics, limitations), comparison, report generation, untested hypotheses and research-section drafting.

Implementation: Python 3.12, Streamlit, the IBM watsonx.ai SDK, **ibm/granite-4-h-small**, pypdf with fontTools, scikit-learn/NumPy and Pydantic. IBM Bob assisted development and troubleshooting. The demonstrated retrieval mode is TF-IDF; optional MiniLM retrieval is disabled.

Inputs and outputs: Up to three PDFs per session, 10 MB and 40 pages per PDF under the configured limits. Outputs include reviewable drafts, page-linked evidence, structured review fields, Markdown, JSON, CSV and BibTeX. The entry point remains **app.py**.

07 Evidence, constraints and responsible use

Local logs report **224 regression tests** and **24 UI helper checks** passing. Five core tasks were reviewed live: question answering, comparison, extraction, summary and report generation, with partial-quality results. Hypothesis and drafting routes were mock-tested, not fully evaluated live.

This internship prototype can produce incorrect citations, omissions, table-reading errors and inaccurate references. Original-source review is required. Credentials remain private. The five-call session guard is not an account-wide spending cap. Local Granite generation was demonstrated; the hosted interface was deployed, but hosted generation was not confirmed.

08 Deliverables and supporting record

Deliverables: the Python source repository, this problem statement, the 19-slide presentation and a descriptive **app.json** manifest. The manifest is documentation, not executable code or a workflow export.

Repository: github.com/SayanGhorai/research-agent
Presentation: ResearchLens_AI_Sayan_Ghorai_Project_Presentation.pptx

Implementation evidence: supplied project code, local logs, screenshots and the 19-slide presentation. No measured productivity gain or answer-accuracy percentage is claimed.